

ValidRange – project description

Background

Two hundred years ago, Alexander Von Humboldt set sail for America for a journey that will lay the foundations for scientific exploration. By describing landscapes, species and their habitats, Von Humboldt sketched a revolutionary global picture of the distribution of life (1). More than two centuries later, and despite an increasingly explored and well-documented natural world, **comprehensive estimates of species' elevational ranges remain surprisingly scarce.**

Understanding species elevational distribution is a **fundamental issue**, first and foremost because mountain regions are characterized by exceptional climatic and topographic heterogeneity, this particularity contributing to their status as **major biodiversity hotspots**, mountains harbouring more than 85% of mammals', amphibians' and birds' species, many of them endemic to their environment (2). Moreover, mountain environments are among the **most vulnerable ecosystems to climate change**, because they experience faster warming than the global average (3). Many of the endemic mountain species are consequently threatened with extinction, but their vulnerability depends both on their range size and on their position along the elevational gradient, emphasizing the importance of accurately documenting their elevational ranges.

Our current limited understanding of species' elevational range across mountains **hampers our ability to predict the effects of global changes**, and consequently to **protect mountain ecosystems under future pressure.**

Aims

The primary aim of the *ValidRange* project is to better understand what shapes species' elevational distribution across mountains. To achieve this goal, we first focus on developing



a globally consistent assessment of species distributions across mountain regions, information that is currently lacking.

The *ValidRange* project is a **global collaborative effort** led by John-Arvid Grytnes within the Norwegian University of Life Sciences (Ås, Norway) and the [Centre for Mountains in Transitions](#) (Univ. of Bergen, Norway), together with the [Mountains in Motion Research Team](#) (Univ. of Bergen, Norway) and the [Global Mountain Biodiversity Assessment](#) (Bern, Switzerland). The project was initiated in 2022 as part of the project “Past, Present, and Future of Alpine Biomes Worldwide” (Mountains in Motion, Univ. of Bergen), and as evidence of the quality and interest of the study, a **first paper and corresponding dataset has been published** in *Nature Communications* (<https://doi.org/10.1038/s41467-026-75210-6>), covering plants’, mammals’, birds’, and reptiles’ taxonomic groups.

Building on this work, *ValidRange* is aiming to **extend and strengthen this dataset**.

Specifically, we are **increasing the scale of the study** by including additional mountain ranges and we are **including the amphibians** as an additional taxonomic group. To achieve this goal, we are reaching out to taxonomic experts around the world.

Tasks

General instructions

As an expert, **your main task would be to help us validate (and correct if necessary), the lower and upper elevational limits** for each species in your taxonomic group and area of expertise.

Please carefully follow the list of instructions in order:

1. **Complete the Google Form** provided in the first introductory email you received from us.





2. **You will receive an email** once you completed the form. **We have unfortunately experienced that many of these emails end up in the spam folder, so, please, check your spam folder!**

3. **Follow the instructions provided in this email.** It guides you towards the dedicated expert-platform and the access to the relevant checklists.

Expert-platform

Once you created your account on the platform, you have a completely free and personal access to a dedicated expert platform. The first thing you will meet on this platform is a map of the mountains of the world based on a revised version of GMBA-maps (4). By clicking on a mountain range and taxonomic group, you will download an excel file with a list of species from the mountain range. See below ([Methods – additional information](#)) for a detailed description of the methods.

Please follow these steps to access the species lists:

1. **Navigate on the map** to the mountain range of your choice.
2. **Select the mountain range by clicking on it.** Icons for the taxonomic groups will appear.
3. **Click on the taxonomic group** of your choice **to download the checklist.**
4. Once it is completed, go back to the platform and **upload the checklist on the same page.**

Specific information on the checklists

Here are the key instructions on how to perform the validation of the checklists:

1. **Validate occurrence:** Please confirm whether the species does occur within the mountain range. If the species **does occur** in that mountain range, **leave blank** the

[presence_corrected] column, if it does **not** occur in that mountain range, enter “no” in *presence_corrected*.

2. Validate elevation data: The elevational limits have been extracted from relevant literature sources (see Table 1). Review (and correct if necessary) the lower and upper elevational limits for each species within the given mountain range. If needed, enter the corrected lower and upper elevation into the designated columns: **[min_corrected]** and **[max_corrected]**. If the current elevation range is correct, indicate this by entering “ok” in the **[validated_elevation_data]** column.

3. Assess your level of confidence: Indicate how confident you are in the input you provide, **on a scale from 1 to 3**. Indicate in the **[confidence_assessment]** column whether you feel **(1)** “high confidence”, **(2)** “medium confidence” or **(3)** “low confidence”.

4. Provide comments: Please add any observations, justifications or notes regarding the assessment of a species’ elevation data in the **[reviewer_comments]** column. This could include information on habitat preferences, uneven distribution patterns, or other factors that might influence a species' data accuracy.

5. Missing species: If you find there are species missing in the spreadsheet, please add them as new rows and add a note “**species added**” to the **[reviewer_comments]** column.

Important note:

You might find that some mountain ranges are quite wide, and that the species’ elevational limits you provide may be non-uniform within this mountain range unit. Some mountain ranges extend over several degrees of latitude and can therefore host very different habitats and climates. If it is the case, please use the **[reviewer_comments]** column for any complementary information on local patterns. It is indeed essential to bear these limitations in mind, and this work lays the foundations for more in-depth validation in the future.

Structure and column naming of the dataset

Dataset columns - fixed:

- **[sciname/genus/family/order]:** The taxonomic ranking according to the chosen taxonomic classification (see Table 1).
- **[seasonal]:** Seasonality code (only for birds and amphibians), based on IUCN mapping standard (5). The code is ranking from 1 to 5: (1) resident; (2) breeding season; (3) non-breeding season; (4) passage; (5) Seasonal occurrence uncertain.
- **[gmba_v2_id]:** Unique identification number for the mountain range.
- **[map_name]:** Standard name of the mountain range.
- **[min_elevation] / [max_elevation]:** The lower and upper elevation limit of the species within the specific mountain range, derived from literature sources (see Table 1).

Expert columns: (see above for specific instructions on how to fill in)

- **[presence_corrected]:** Confirmation of the presence of a species in the assigned mountain range.
- **[min_corrected] and [max_corrected]:** Correction of the species' elevational limits.
- **[validated_elevation_data]:** Enter “ok” if the specified elevational range estimate is correct.
- **[confidence_assessment]:** Level of confidence on your assessment for each species on a scale from **(1)** “high confidence”, to **(3)** “low confidence”.
- **[reviewer_comments]:** Additional comments about the distribution, elevational range, or habitat preferences of the species.

Collaboration

By collaborating in this project, you are **guaranteed to be included as a co-author of the publication(s)** we are planning on writing. You would also become **part of a global network of regional experts** connected to an international, cross-disciplinary platform

dedicated to the assessment, conservation, and sustainable management of mountain biodiversity, coordinated by the [Global Mountain Biodiversity Assessment](#).

Methods – additional information

We have compiled elevational ranges of mountain species based on literature and spatial analysis. For the mountain delineations, we used the GMBA Mountain Inventory v2.0 (4), specifically the “300 selection”, a subset of 291 non-overlapping major mountain systems and ranges. For the species’ ranges and elevational limits, we based our work on several sources detailed below (Table 1).

We selected species with a distribution range overlapping mountains areas of at least 10 km².

GROUP	ELEVATIONAL RANGES	DISTRIBUTION RANGES
Mammals	Obtained from the machine-readable version of the <i>Handbook of Mammals of the World</i> (HMW) (6)	Mammal Diversity Database (7)
Reptiles	SquamBase—A database of squamate (Reptilia: Squamata) traits (8)	Global Assessment of Reptile Distribution (GARD) (15, 16)
Birds	Quintero & Jetz (11)	BirdLife International and Handbook of the Birds of the World (12)

Amphibians	Guirguis et al. (13)	IUCN SSC Amphibian Specialist Group. 2023. <i>State of the World's Amphibians. The Second Global Amphibian Assessment</i> . (5)
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Table 1: Sources information for elevational limits and distribution of mammals, reptiles, birds and amphibians.

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